Abhishek Kundu

Email: **abhishektaurian@gmail.com** | Phone: **+91 8890430564** | Languages: English, Hindi | Address: Roopena Agrahara, Bangalore-560068

Lead Data Engineer

Seeking a position in an organization that provides me ample opportunity to explore & excel while carving out the niche for organizational, professional as well as personal goals.

Summary of Skills

- Result oriented IT professional with 9+ years of experience in Data Engineering with considerable knowledge in the analysis, design, coding, maintaining and re-engineering applications; currently working with HTC Global Services, Bangalore.
- Successful track record of implementing various projects within agreed cost/timelines; possess extensive knowledge in ETL, Batch & Real time big data processing, Data warehousing, ETL Framework development and Application development using Hadoop and cloud tools such as Azure and AWS; possess good knowledge in improving the performance, resolving small file problems and optimization of the existing algorithms in Hadoop using Spark Session, Spark SQL, Data Frame, and developed HQL scripts using Spark jars for handling different data sets; Handled file types such as CSV, txt, parquet, JSON
- In-depth knowledge of all phases of Software Development Life Cycle (SDLC) with concentration on Design, Development & Implementation.
- Experience of working in agile methodology with focus on completion of tasks in stipulated timelines
- Designed ETL Frameworks for the developers to use it for the data processing in their own project.
- Worked with different customers, interacted with them, understood their requirements, developed end to end pipelines and provided sufficient knowledge transfer for the support teams.

Professional Experience

- HTC Global Services (Insurance), Bangalore ♦ Lead Data Engineer ♦ May-22-Dec24
- KPI Partners (Healthcare), Bangalore ♦ Lead Data Engineer♦ Mar'21-May'22
- LTIMindtree (Healthcare), Bangalore ♦ Specialist-cloud data engineer ♦ Mar'20-Mar-21
- Datamatics Global Services Limited (Retail), Bangalore ♦ Consultant ♦ July 19-Mar'20
- Capgemini Technology Services (Retail), Bangalore ♦ Consultant ♦ Dec'15-June 19

Education

- Bachelor's Degree (B.Tech in Electronics and Communication)
  Rajasthan Technical University (2010–2014)

Technical Experience

Big Data Ecosystem (ETL & Data Engineering):

Hadoop(Cloudera & Hortonworks): HDFS, YARN, Hive, Spark, Kafka

Cloud Tools:

- **Microsoft Azure:** Azure Data Factory
- **AWS**: S3, Lambda, SNS, SQS, Cloud Watch & KMS, Glue, EMR, Redshift, Redshift Spectrum, DynamoDB, RDS, Aurora, Athena, API Gateway, Route 53, ALB, DMS, Code Pipeline, Code Repo, Secrets Manager, Step Function, IAM
- **Databricks:** Databricks cloud platform

Programming Languages : SQL, Python, Scala, Core Java & Unix

Databases : SQL Server, DB2, ORACLE, Postgres

Replicate Tool : Qlik Replicate, Apache Nifi

Development Tools : Maven, Eclipse, Visual code, PyCharm, IntelliJ & Notebooks

Version Control : GitLab, Bitbucket, Github

Scheduling & Orchestration : Autosys, Databricks cloud, cron job expression

Data Science Libraries : NumPy, Pandas, scikit-learn, Tensorflow

Project Details

Project #1	**ETL/Data Migration/Cloud Infrastructure Creation/Data Analytics/Devops**
Client	Statefarm
Technology	Python, Pytest, Terraform, BDD, Terratest, Qlik Replicate, GitLab CI/CD, AWS (ALB, Lambda, Secret Manager, IAM, CloudWatch, API Gateway, Route53, RDS, Aurora)
Description	This project involved migration of historical data from the Mainframe DB2 system to AWS RDS using Qlik replicate tool. Later this Qlik tool would be removed upon deploying the legacy application directly into AWS cloud and connecting the external API's to AWS RDS. This data would further be used for integrating with different datasets and use it for analytics purposes.

Role	● Built AWS cloud infrastructure like creation of RDS, API Gateway, ALB, Secrets manager using Terraform scripts ● Integrated AWS Cloud with Qlik Replicate. ● Created Qlik Replicate jobs for data migration ● Created CI/CD pipeline for production deployment. ● Written Pytest codes for the prebuilt API's.

Project #2	**API Creation/ BDD testing/ Automation testing/Data Analytics/Application Development**
Client	Statefarm
Technology	Python, Pytest, BDD, Terratest, GitLab CI/CD, AWS (ALB, Lambda, API Gateway, Route53)
Description	This project involved creation of Custom Rest API's for different applications. These API's would be used by the sales and marketing team to record their data in the AWS RDS. This project also involved creation of all the AWS resources required and integration with Vault.
Role	● Designed and developed custom APIs in Python. ● Wrote Pytest cases for all APIs ● Implemented the BDD scenarios.

Project #3	**ETL Framework Development/ETL Application Development/Data Analytics/Architect**
Client	Thermofisher Scientific
Technology	Python, Spark, Databricks, AWS (S3, Athena, Redshift Spectrum, Lambda), GitHub, Jenkins, SQL
Description	This project involved migration of data from the SAP systems to the AWS cloud environment. The business logics were written for the legacy SAP systems which

	were outdated and joining different tables and doing certain transformations were difficult to achieve. These business logics were converted into sql queries and placed in the S3 buckets. The Sql Framework built would pull these queries from S3 bucket and run those transformations on Databricks cloud and put the resultant data into processed S3 bucket.
Role	● Created a SQL framework to pull the queries from S3 buckets. These queries would run on Databricks cloud and give the resultant data in S3 ● Automated the data loads into S3 and exposed it in Athena using Table creation Framework. ● These frameworks would work for entire enterprise datasets and all different project use cases based on need. Just a few project related parameters needed to be passed in the framework.

Project #4	**ETL Framework Development/ETL Application Development/Data Analytics/Architect**
Client	Thermofisher Scientific
Technology	Python, Spark, Databricks, AWS (S3, Athena, Redshift Spectrum, Lambda), GitHub, Jenkins, SQL
Description	This project involved creation of Flat File Framework which would transform structured and semi structured data put by different vendors into S3 buckets. As soon as the data drops on S3, the Lambda would trigger the Databricks job and the transformation and loading of data starts. This framework was used by multiple projects. All the developers only had to put a few project related parameters and the data processing would be done.
Role	● Built a flat file framework to process vendor-provided files in S3. ● Automated data exposure to Redshift Spectrum and Athena using Table Creation Framework. ● Also created Schema Registry Framework which would register the schemas for different tables automatically whenever we put updated schemas in csv format in the specified S3 bucket.

Project #5	**ETL/Data Migration/Data Transformation/Data Analytics/Architect**
Client	Thermofisher Scientific
Technology	Python, Spark, Databricks, AWS (S3, Athena, Redshift Spectrum, Lambda), GitHub, Jenkins, SQL
Description	This project involved creation of databricks job which would fetch data from different source systems and do the necessary transformation and put it on S3. The entire ETL process would be done using the databricks job.
Role	● Developed Databricks jobs for processing structured and semi-structured data from source systems using Rest Api and Bulk Api ● Automated data exposure to Athena and Redshift Spectrum using Table Creation Framework.

Project #6	**ETL/Data Migration/Data Pipeline/Data Analytics**
Client	Resmed
Technology	Python, Pytest, Spark, AWS (Glue, S3, Lambda, EMR, CloudFormation, DynamoDB, CodePipeline, CodeCommit, Athena, EC2), SQL
Description	Data coming from multiple healthcare devices had to be extracted from sql server and processed on AWS cloud for analytical purposes.
Role	● Extracted data from MySQL to S3 using DMS and processed it with AWS Glue and EMR. ● Used Step Functions for orchestration and CloudFormation for deployments.

Project #7	**Framework Development/Application Development/Data Analytics/Architect**
Client	Resmed
Technology	Python, Pytest, Spark, AWS (Glue, S3, Lambda, EMR, CloudFormation, DynamoDB, CodePipeline, CodeCommit, Athena, EC2), SQL
Description	Once the entire pipeline runs and the data has been stored in target, we need to check if the data is clean or not or if there is any leakage in the data processing. We also needed to check the count of data and the proper masking of data was happening. Hence built the Data Validation Framework.
Role	● Developed a data validation framework for verifying bucket data consistency. This would compare the source and target and give the result if the data is fit enough for usage or if there is any data leakage ● Also worked in Pytest cases of some of the existing Glue codes. ● Was also involved in optimizing current pipelines and resolving older issues.

Project #8	**ETL/ Data Migration/ Data Analytics**
Client	NTT Data/Honeywell
Technology	Python, Spark, AWS (Glue, S3, Redshift Spectrum, Lambda, CloudFormation, SNS, SQS, DLQ, CodePipeline, Systems Manager), SQL
Description	This project involved data migration from HANA DB to AWS cloud environment. The dataset needed to join with other datasets in order to extract meaningful information out of it.
Role	● Extracted data from HANA DB using Informatica and transformed it with AWS Glue. ● Used Python and Spark to transform and load the data in S3. ● Deployed code to higher environments using TFS CI/CD pipeline.

Project #9	**Data Warehousing/Data Migration/ETL/Data Analytics**
Client	Coca-Cola
Technology	Python, Spark, Apache NiFi, AWS (Glue, EMR, S3, Athena), Autosys, Shell Scripting, SQL
Description	This project involved data migration from SQL to AWS S3 using Apache Nifi. Once the extraction is done, AWS Glue would transform the data and put the transformed data into a processed bucket. The data is exposed to Athena. This data would further be used by microstrategy to visualize the data in the form of graphs and tables.
Role	● Automated data pipelines using Apache NiFi and AWS Glue.. ● Managed scheduling with Autosys and shell scripting.

Project #10	**Data Warehousing/Data Migration/ETL/Data Analytics**
Client	Coca-Cola
Technology	Python, Spark, Apache NiFi, AWS (Glue, EMR, S3, Athena), Autosys, Shell Scripting, SQL
Description	This project involved data migration from SQL and Oracle to AWS S3 using Apache Nifi. Once the extraction is done, AWS EMR would transform the data and put the transformed data into a processed bucket. The data is exposed to Athena. This data would further be used by microstrategy to visualize the data in the form of graphs and tables.
Role	● Worked as a developer to enhance the already built pipeline here. ● Worked for optimizing the pipeline and correcting the business logic in a team effort. ● Was also a part of the support team and looked into the existing issues and enhancing the data pipeline.